HANDSET REV A / SYSTEM SCHEMATIC INDEX

Engineering draft. Retained ESP32-S3 / MakerFocus 3000 mAh / external display / all peripheral interfaces.

ESP32 host, boot and reset

File: core.kicad_sch

Key matrix and trackball

File: keys.kicad_sch

USB-C data and ESD

File: usb.kicad_sch

Fused USB input and eFuse

File: usb-input.kicad_sch

USB-C source detection and permission

File: usb-detect.kicad_sch

Battery, thermistor and programmable charger

File: power.kicad_sch

Power control and fuel gauge

File: power-control.kicad_sch

Main 3.3 V buck-boost supply

File: reg3.kicad_sch

Auxiliary 5 V and 1.8 V supplies

File: aux-power.kicad_sch

Regulated modem supply and enable

File: modem-power.kicad_sch

SX1262 LoRa radio

File: radios.kicad_sch

CC1101 sub-GHz radio

File: cc-radio.kicad_sch

125 kHz reader analog circuit

File: lf-radio.kicad_sch

LF reader logic translation

File: lf-support.kicad_sch

GNSS receiver and passive antenna

File: gnss-antennas.kicad_sch

NFC reader, oscillator and local supplies

File: nfc.kicad_sch

Media speaker amplifier and digital microphone

File: audio.kicad_sch

Cellular adapter and call-audio harnesses

File: interfaces.kicad_sch

Protected external module port

File: protection.kicad_sch

MicroSD and case-mounted infrared

File: storage-ir.kicad_sch

External display, touch and backlight

File: display.kicad_sch

Always-on charging supervisor

File: power-supervisor.kicad_sch

Display supply and translation

File: display-interface.kicad_sch

Expansion control and isolation

File: expansion-control.kicad_sch

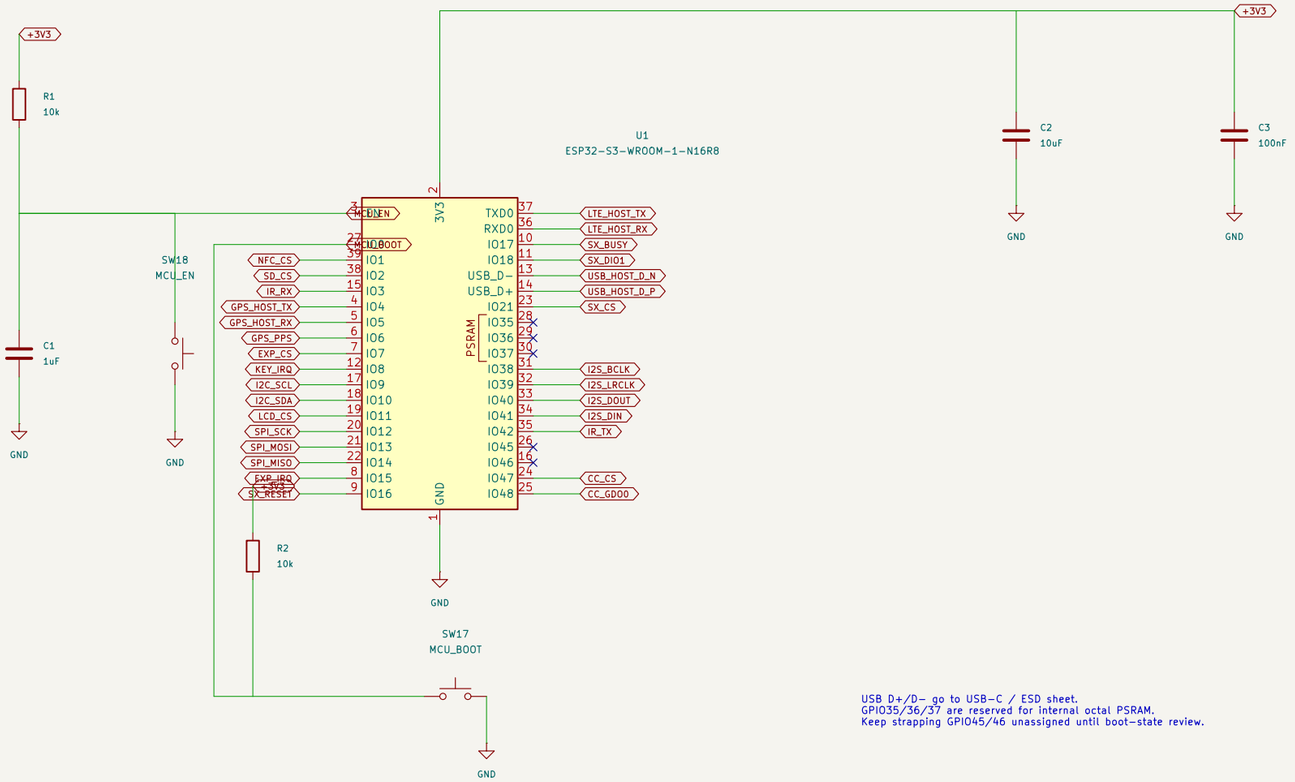
Sheet: /
File: handset.kicad_sch

Title:

Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/25

ESP32 host, boot and reset

Retain N16R8. Module pads 28/29/30 belong to octal PSRAM. USB programming uses the USB sheet.



RESET / EN: 10k pull-up + 1uF
BOOT held low during reset selects download mode.

USB D+/D- go to USB-C / ESD sheet.
GPIO35/36/37 are reserved for internal octal PSRAM.
Keep strapping GPIO45/46 unassigned until boot-state review.

Engineering draft / named nets are common / NC = intentional unused pin

Circuit review in progress — not released for fabrication

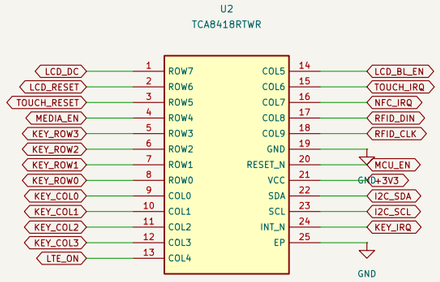
Sheet: /ESP32 host, boot and reset/
File: core.kicad_sch

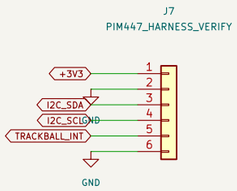
Title: ESP32 host, boot and reset

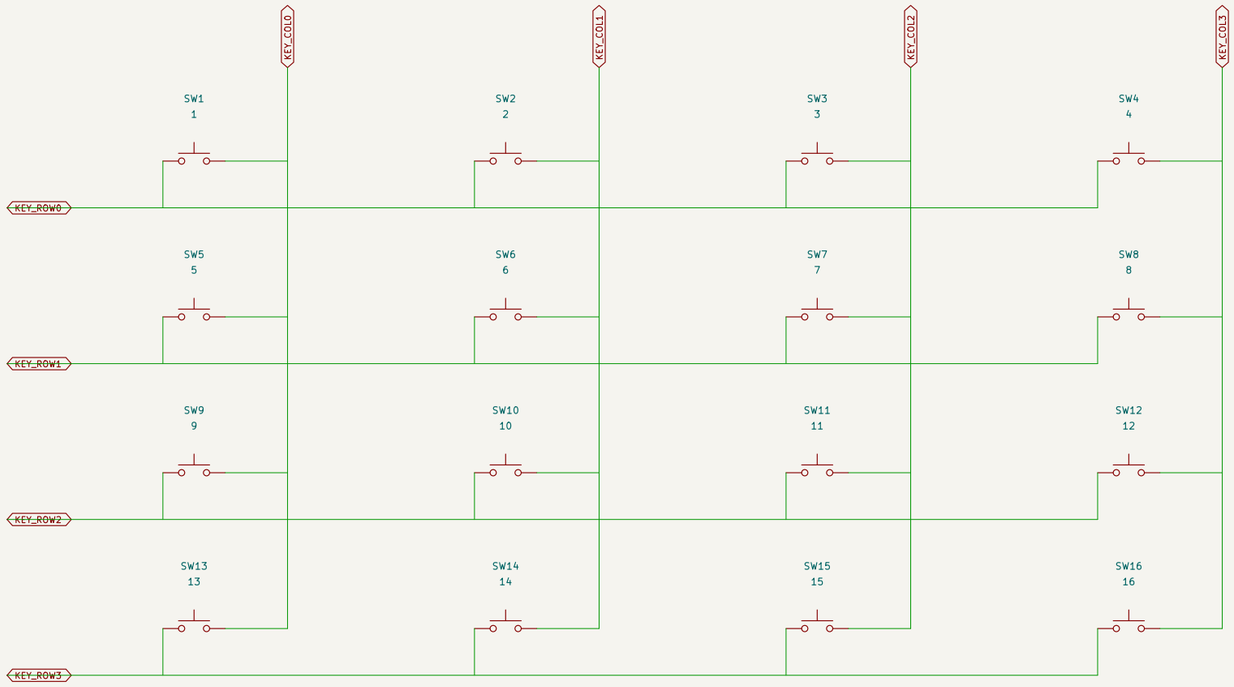
Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 2/25

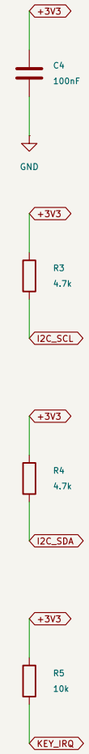
Key matrix and trackball

TCA8418: 4 rows x 4 columns; remaining pins are firmware-configured GPIO. Confirm PIM447 harness orientation.









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Circuit review in progress — not released for fabrication

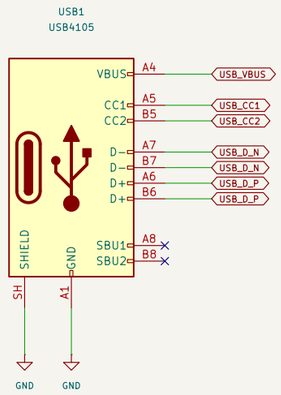
Sheet: /Key matrix and trackball/
File: keys.kicad_sch

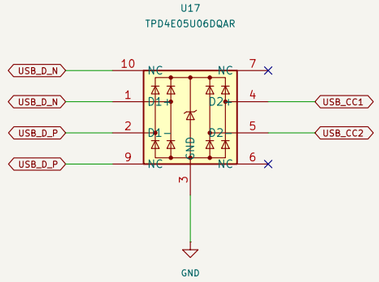
Title: Key matrix and trackball

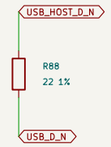
Size: A3 | Date: 2026-09-24 | **Rev: A — engineering draft**
KiCad E.D.A. 10.0.6 | Id: 3/25

USB-C data and ESD

U17 NC lands 10/9 externally bridge to 1/2 for pass-through routing; no internal connection. R88/R89 host series termination, initial 22 ohms; qualify on hardware.









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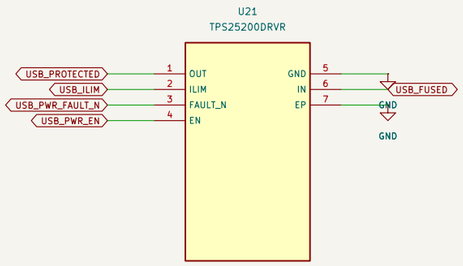
Sheet: /USB-C data and ESD/
File: usb.kicad_sch

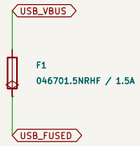
Title: USB-C data and ESD

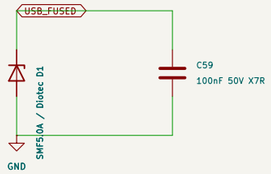
Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
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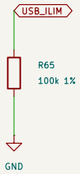
Fused USB input and eFuse

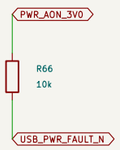
TPS25200 R_ILIM = 100k, 1%. F1/TVS pulse coordination requires bench validation. No USB PD negotiation.

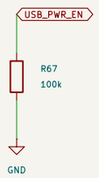












Supply flag: external USB VBUS through F1. No internal source is implied.

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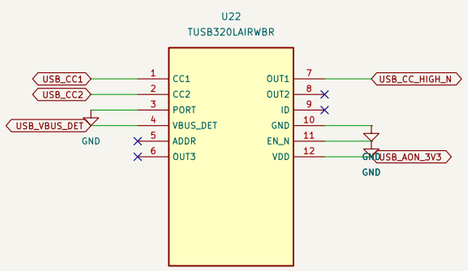
Sheet: /Fused USB input and eFuse/
File: usb-input.kicad_sch

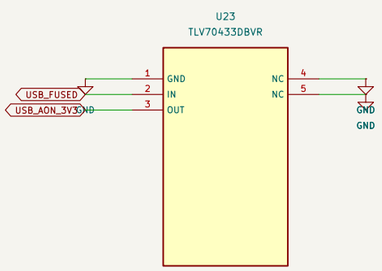
Title: Fused USB input and eFuse

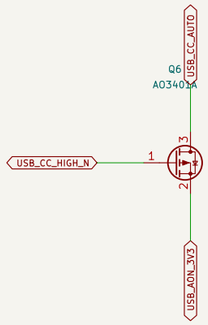
Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 5/25

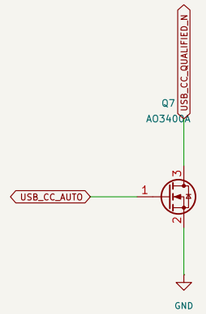
USB-C source detection and permission

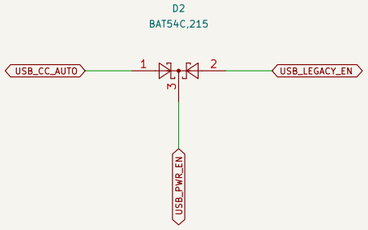
Sink-only internal Rd. OUT1 low authorizes Type-C 1.5A/3A. Legacy permission requires USB enumeration; see charger-policy.json.

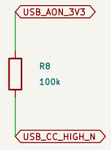


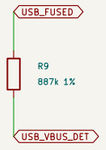


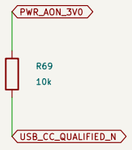


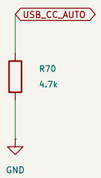


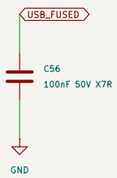


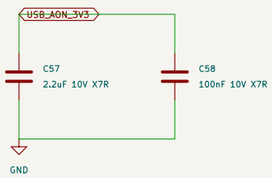












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Circuit review in progress — not released for fabrication

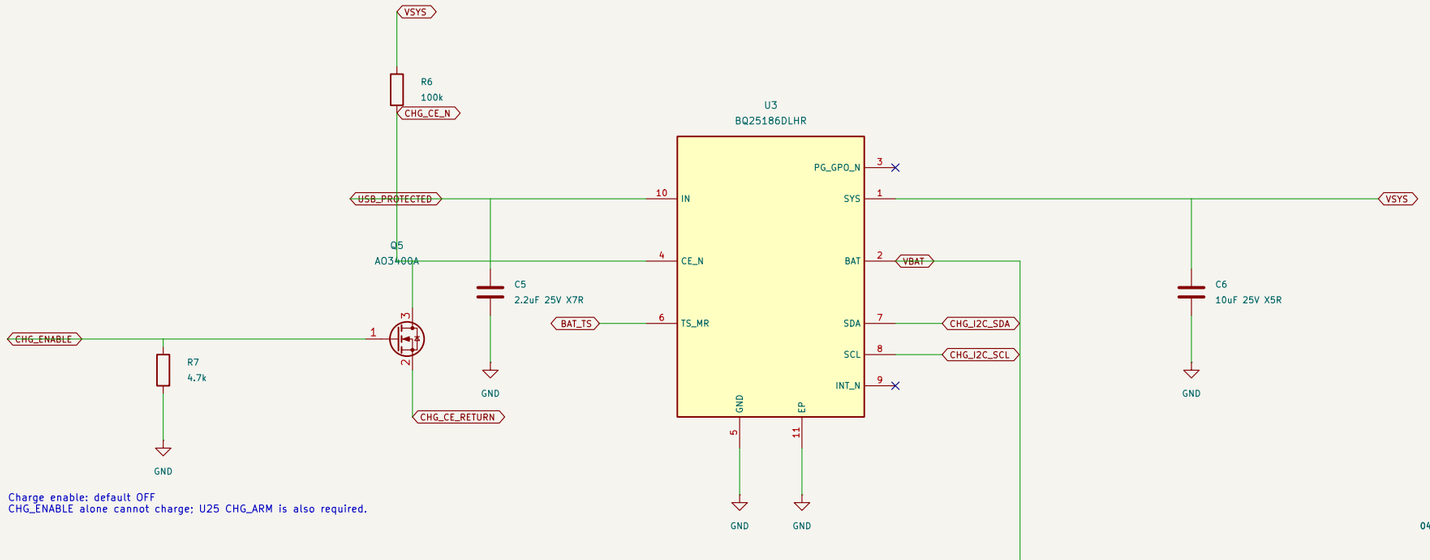
Sheet: /USB-C source detection and permission/
File: usb-detect.kicad_sch

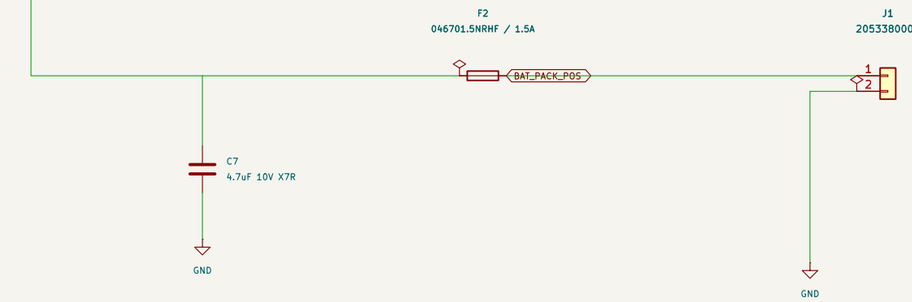
Title: USB-C source detection and permission

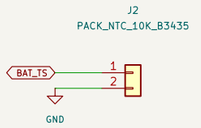
Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 6/25

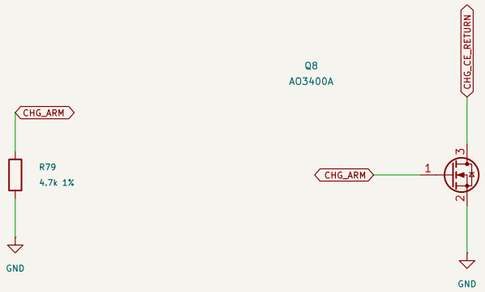
Battery, thermistor and programmable charger

Verify actual battery polarity. J2 is a battery-mounted 10k B3435 NTC. CE defaults OFF. Firmware must apply charger-policy.json.









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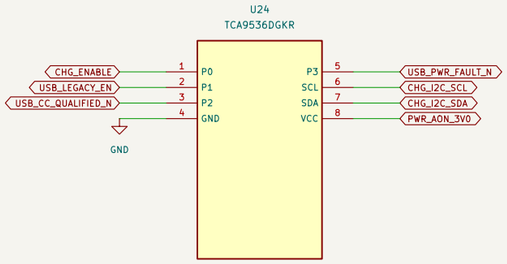
Sheet: /Battery, thermistor and programmable charger/
File: power.kicad_sch

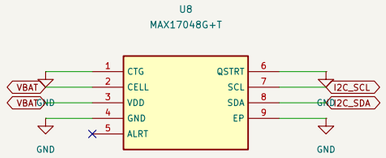
Title: Battery, thermistor and programmable charger

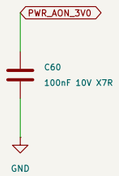
Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 7/25

Power control and fuel gauge

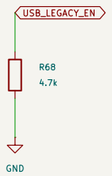
I2C: TCA9536 0x41, MAX17048 0x36, charger 0x6A. Preload low output latches before enabling charger-control outputs.











Engineering draft / named nets are common / NC = intentional unused pin

Circuit review in progress — not released for fabrication

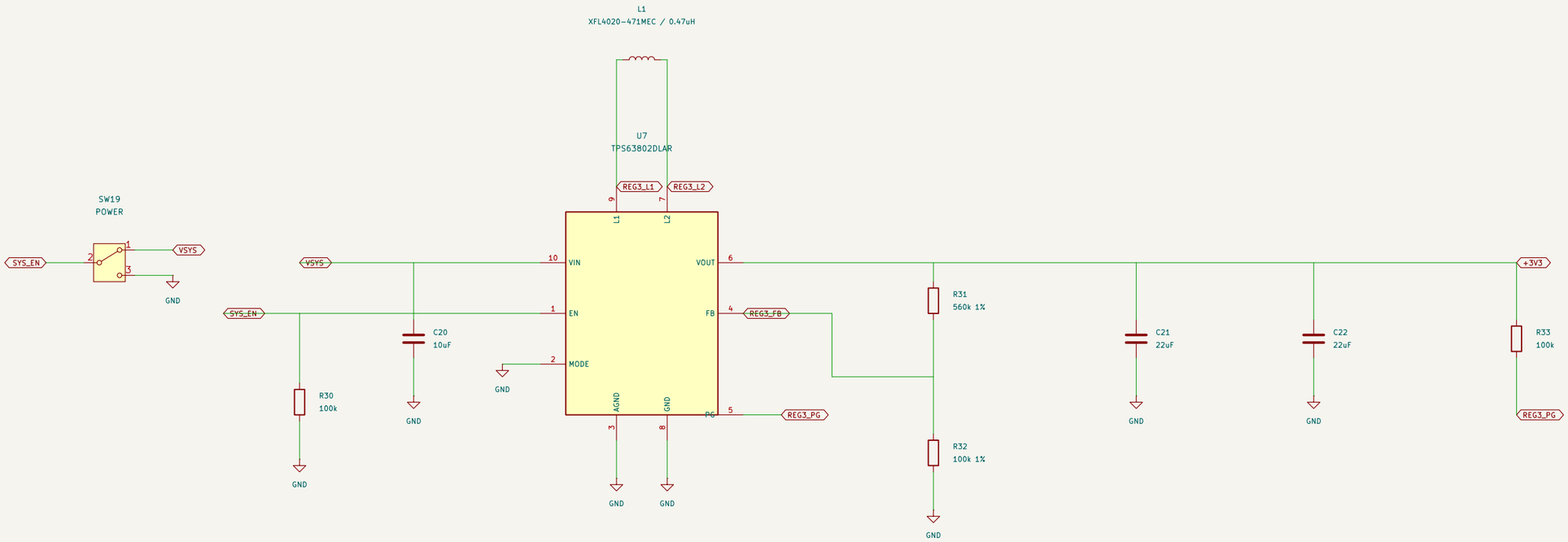
Sheet: /Power control and fuel gauge/
File: power-control.kicad_sch

Title: Power control and fuel gauge

Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 8/25

Main 3.3 V buck–boost supply

TPS63802 local PCB copper is routed. External distribution, effective capacitance, thermal limits and load transients remain to qualify.



SW19 controls SYS_EN.
Battery / charger circuitry can remain energized while logic rails are off.

0.6 V reference x (1 + 560k / 100k) = 3.3 V nominal.
Feedback senses the output capacitor node: keep it clear of the switching loop.

Engineering draft / named nets are common / NC = intentional unused pin

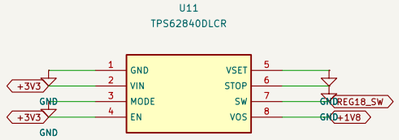
Circuit review in progress – not released for fabrication

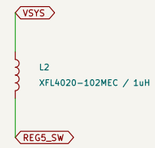
Sheet: /Main 3.3 V buck–boost supply/		
File: reg3.kicad_sch		
Title: Main 3.3 V buck–boost supply		
Size: A3	Date: 2026-09-24	**Rev: A – engineering draft**
KiCad E.D.A. 10.0.6		Id: 9/25

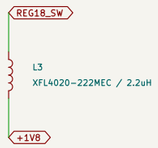
Auxiliary 5 V and 1.8 V supplies

Switching loops must be short in PCB layout. The 5 V rail also supplies display backlight and the LF reader.

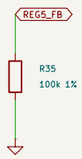








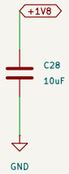












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Circuit review in progress — not released for fabrication

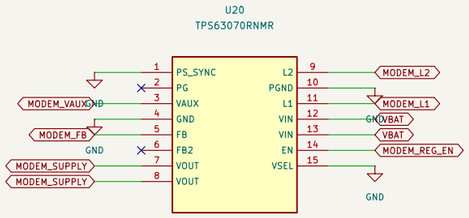
Sheet: /Auxiliary 5 V and 1.8 V supplies/
File: aux-power.kicad_sch

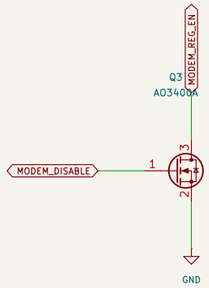
Title: Auxiliary 5 V and 1.8 V supplies

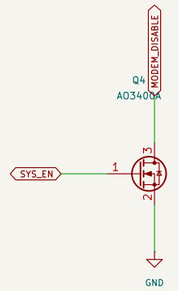
Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 10/25

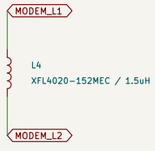
Regulated modem supply and enable

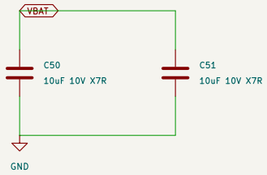
Nominal output 3.792 V. Exact modem burst current and MakerFocus pack current budget are unresolved release blockers.

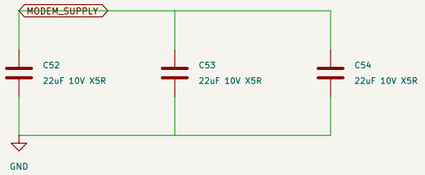




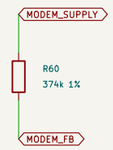


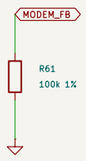


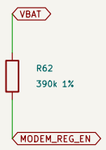


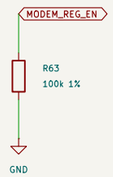


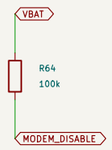












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Circuit review in progress — not released for fabrication

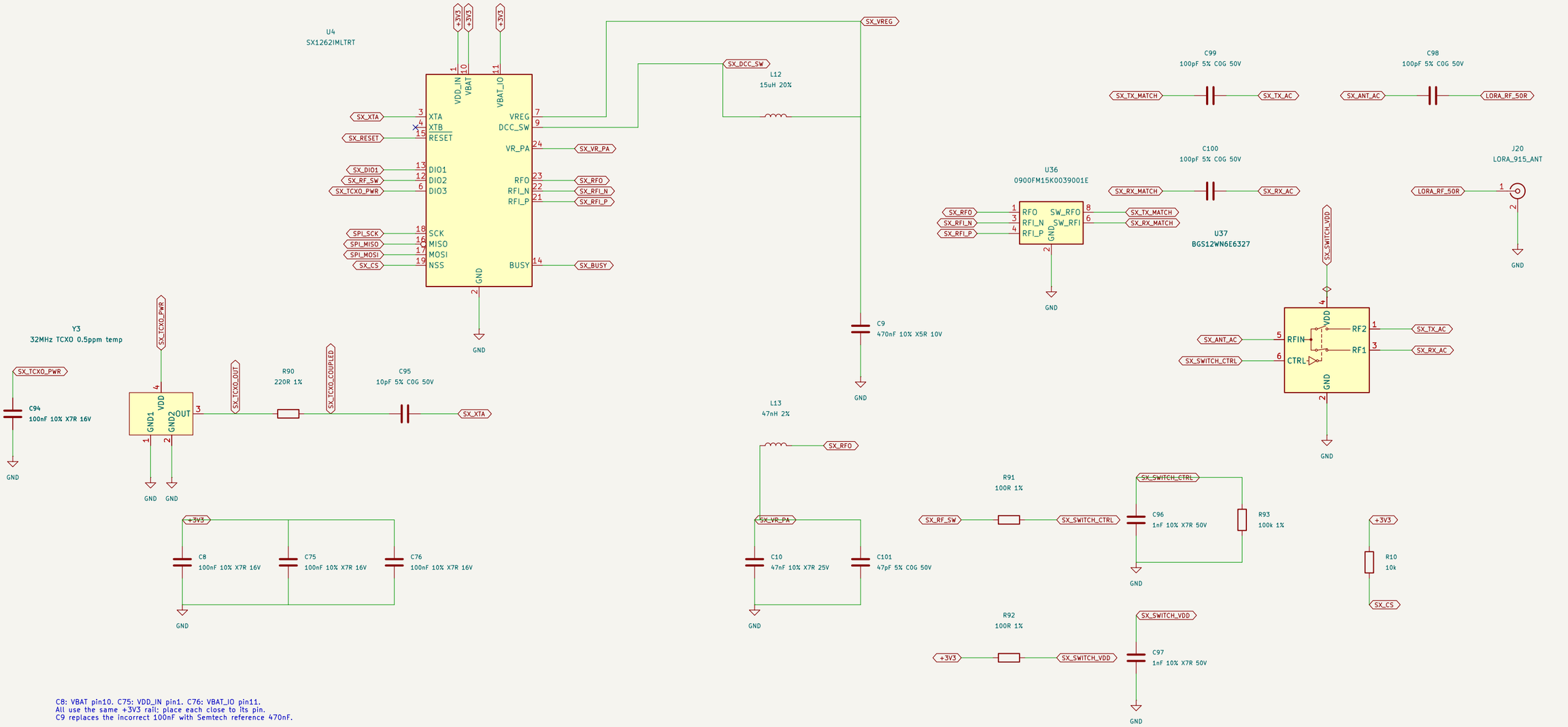
Sheet: /Regulated modem supply and enable/
File: modem-power.kicad_sch

Title: Regulated modem supply and enable

Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 11/25

SX1262 LoRa radio

SX1262 clock, supply, PA feed and RF matching/switch captured. RF layout incomplete; do not power.



C8: VBAT pin10. C75: VDD_IN pin1. C76: VBAT_IO pin11.
All use the same +3V3 rail; place each close to its pin.
C9 replaces the incorrect 100nF with Semtech reference 470nF.

Y3 pads 1/2 = GND, 3 = OUT, 4 = VDD. XTB is intentionally NC.
Configure DIO3 for 1.8V; provisional startup allowance 5ms.
Confirm TCXO output <=1.2Vpp; no qualified maximum in series data.

U37: CTRL=0 selects RF1/RX; CTRL=1 selects RF2/TX.
R93 defaults to RX. C98/C99/C100 explicitly block DC.
100pF RF blocks are starting values, subject to RF measurement.
No RF route or thermal/ground-via qualification in this checkpoint.

SetRegulatorMode(0x01) only in STDBY_RC, after hardware qualification.
L12 supplies the core, not the SX1262 high-power PA.
L13: 47nH PA choke; C10/C101: 47nF/47pF bypass.
RAK4270 values; final RF layout and qualification remain open.

Engineering draft / named nets are common / NC = intentional unused pin

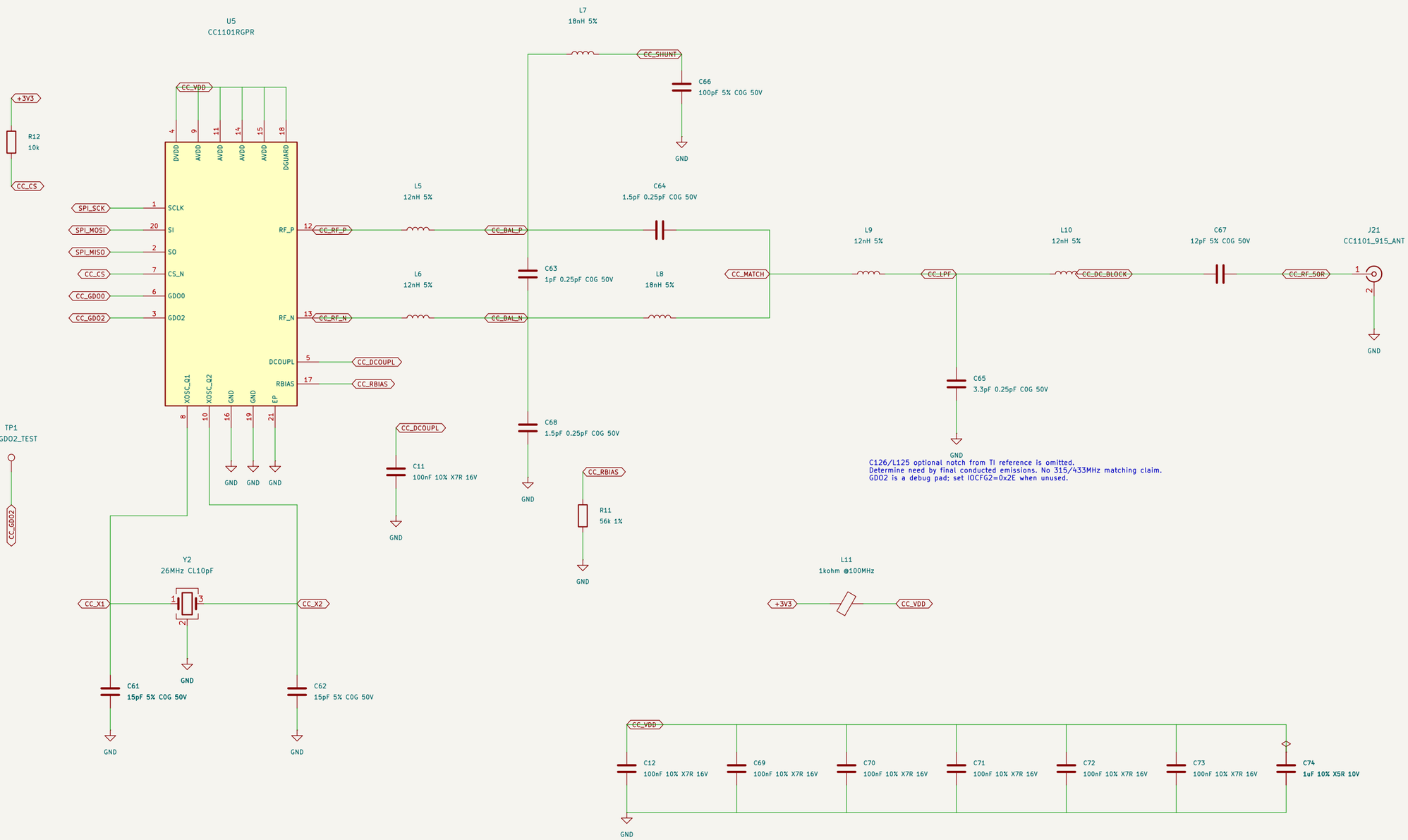
Circuit review in progress — not released for fabrication

Sheet: /SX1262 LoRa radio/	
File: radios.kicad_sch	
Title: SX1262 LoRa radio	
Size: A3	Date: 2026-10-08
KiCad E.D.A. 10.0.6	Id: 12/25

Rev: A — engineering draft

CC1101 sub-GHz radio

868/915MHz frontend: TI SWRR045 rev3.0 / SWRS061I Fig11. RF layout and tuning are NOT qualified. Separate 50-ohm antenna.



C126/L125 optional notch from TI reference is omitted.
Determine need by final conducted emissions. No 315/433MHz matching claim.
GDO2 is a debug pad; set IOCFG2=0x2E when unused.

Y2: ABM8-26.000MHZ-10-D-1-G-T
CL = 15pF / 2 + estimated 2.5pF stray = 10pF.
Measure oscillator frequency, startup and drive level.

CC_VDD flag: main +3V3 through passive L11; not an independent source.
Place one 100nF at each supply pin group, plus local 1uF bulk.

Engineering draft / named nets are common / NC = intentional unused pin

Circuit review in progress — not released for fabrication

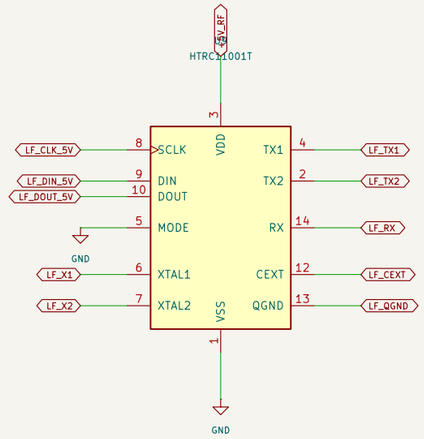
Sheet: /CC1101 sub-GHz radio/
File: cc-radio.kicad_sch

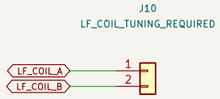
Title: CC1101 sub-GHz radio

Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 13/25

125 kHz reader analog circuit

HTRC110 requires its oscillator, QGND/CEXT bypass, resonant coil network and a timing-capable host interface.







Engineering draft / named nets are common / NC = intentional unused pin

Circuit review in progress — not released for fabrication

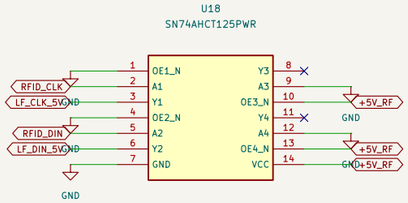
Sheet: /125 kHz reader analog circuit/
File: lf-radio.kicad_sch

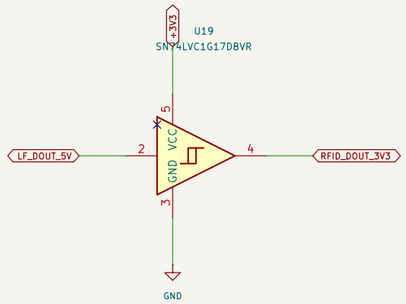
Title: 125 kHz reader analog circuit

Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 14/25

LF reader logic translation

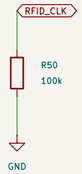
AHCT inputs accept 3.3 V logic at 5 V supply; LVC return buffer is powered at 3.3 V. Host timing remains a design requirement.

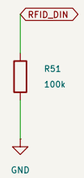












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Circuit review in progress — not released for fabrication

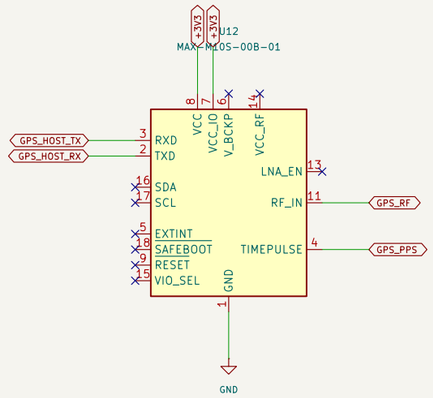
Sheet: /LF reader logic translation/
File: lf-support.kicad_sch

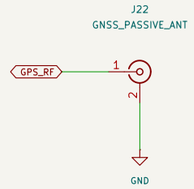
Title: LF reader logic translation

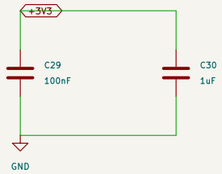
Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 15/25

GNSS receiver and passive antenna

MAX-M10S passive antenna. V_BCKP remains open: cold starts after power removal. No direct LiPo connection.







Engineering draft / named nets are common / NC = intentional unused pin

Circuit review in progress — not released for fabrication

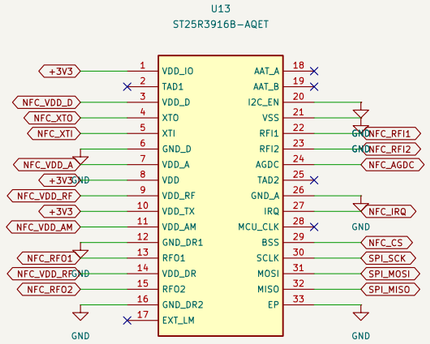
Sheet: /GNSS receiver and passive antenna/
File: gnss-antennas.kicad_sch

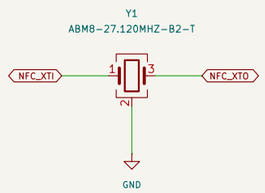
Title: GNSS receiver and passive antenna

Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 16/25

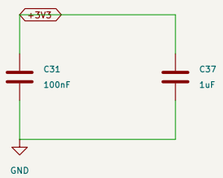
NFC reader, oscillator and local supplies

ST25R3916B needs a tuned case-mounted loop and TX/RX matching network. Verify crystal load against the selected crystal.





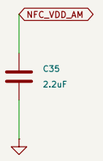


















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Circuit review in progress — not released for fabrication

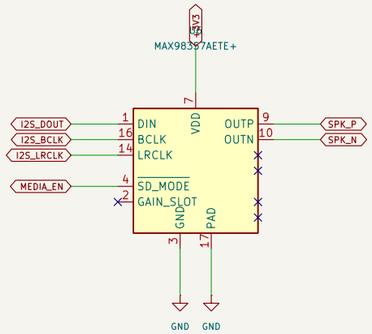
Sheet: /NFC reader, oscillator and local supplies/
File: nfc.kicad_sch

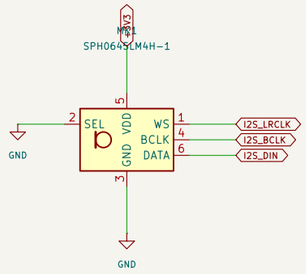
Title: NFC reader, oscillator and local supplies

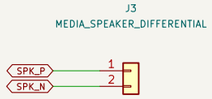
Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 17/25

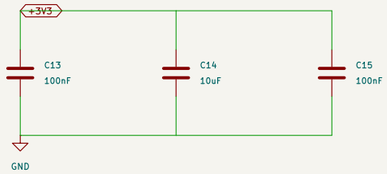
Media speaker amplifier and digital microphone

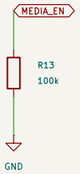
Top-mounted media speaker uses differential outputs: neither speaker wire is ground. Digital microphone is case-bottom mounted.











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Circuit review in progress — not released for fabrication

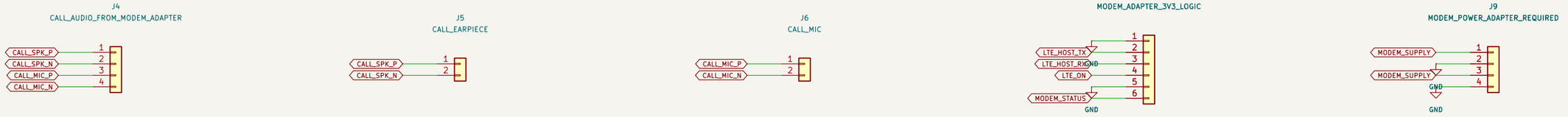
Sheet: /Media speaker amplifier and digital microphone/
File: audio.kicad_sch

Title: Media speaker amplifier and digital microphone

Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 18/25

Cellular adapter and call-audio harnesses

Replaceable modem adapter owns SIM, cellular RF and modem-level translation. No universal-carrier guarantee. Speaker top / mic bottom.



Engineering draft / named nets are common / NC = intentional unused pin

Circuit review in progress — not released for fabrication

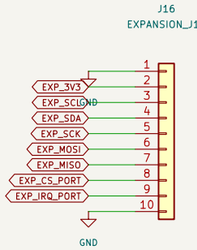
Sheet: /Cellular adapter and call-audio harnesses/
File: interfaces.kicad_sch

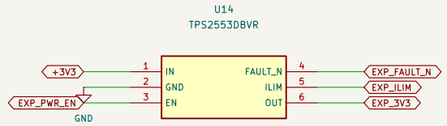
Title: Cellular adapter and call-audio harnesses

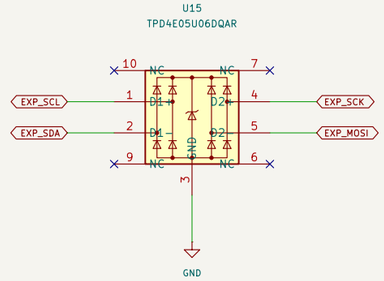
Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 19/25

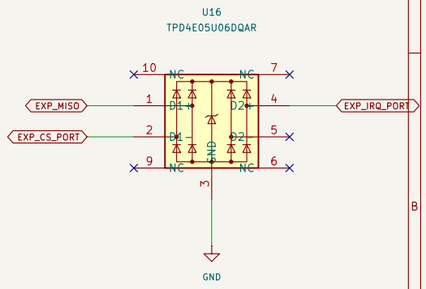
Protected external module port

Port signals pass through U34 on expansion-control. U14 EN defaults low; fault drives U33 and the hardware signal interlock.
No hot-plug approval.





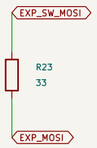




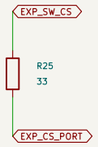




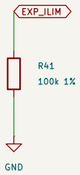


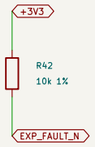


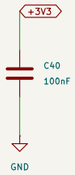














Engineering draft / named nets are common / NC = intentional unused pin

Circuit review in progress — not released for fabrication

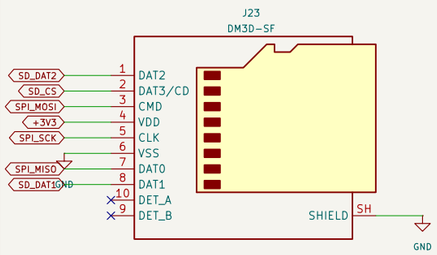
Sheet: /Protected external module port/
File: protection.kicad_sch

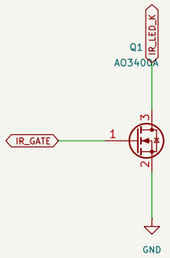
Title: Protected external module port

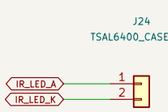
Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 20/25

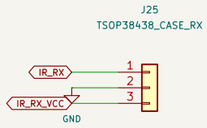
MicroSD and case-mounted infrared

SPI microSD, transistor-driven IR LED, filtered receiver supply. Confirm case LED/receiver polarity and GPIO3 boot behavior.







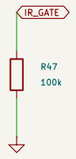


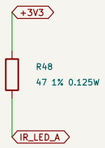




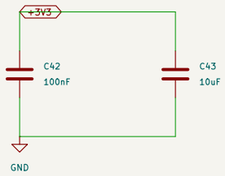














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Circuit review in progress — not released for fabrication

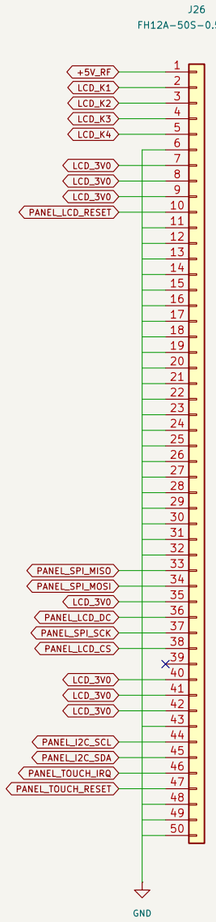
Sheet: /MicroSD and case-mounted infrared/
File: storage-ir.kicad_sch

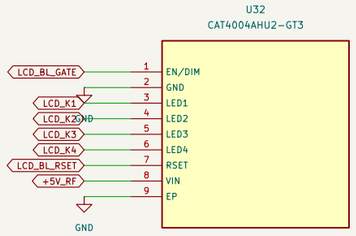
Title: MicroSD and case-mounted infrared

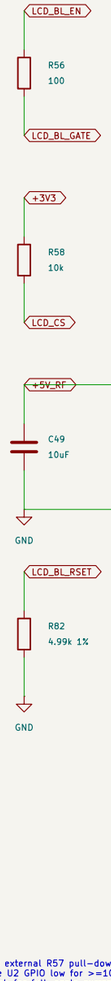
Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 21/25

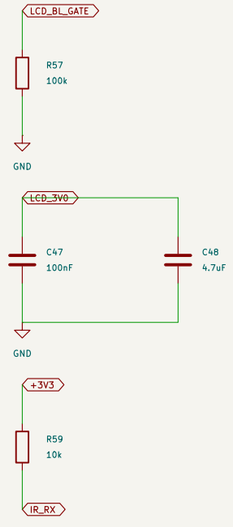
External display, touch and backlight

CAT4004A: four current sinks, 4.99k sets approx. 15mA/channel. On/off only until dimming timing is qualified. Do not fabricate.









EN/DIM: external R57 pull-down retains default off.
Initialize U2 GPIO low for >=10ms before enabling.
Keep high for full-scale current; do not apply ordinary PWM.
RSET gain, current drift, heat and purchased panel remain unqualified.

Engineering draft / named nets are common / NC = intentional unused pin

Circuit review in progress — not released for fabrication

Sheet: /External display, touch and backlight/
File: display.kicad_sch

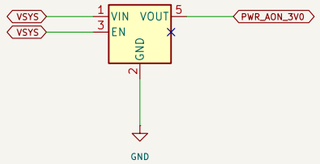
Title: External display, touch and backlight

Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 22/25

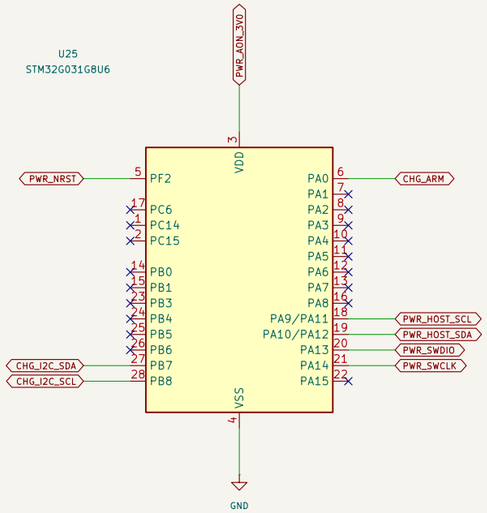
Always-on charging supervisor

Separate supervisor selected. U3 remains the charger. Main host powers off at SW19. Circuit capture only; routing, STM32 firmware and fault qualification pending.

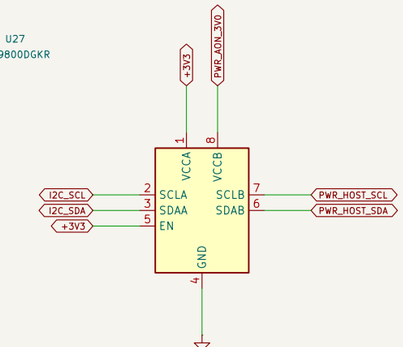
U26
TPS7A0230PDBVR



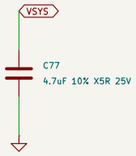
U25
STM32G031G8U6

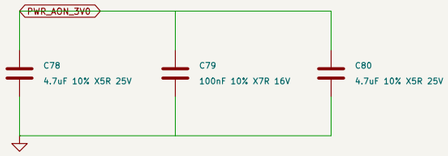


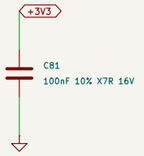
U27
TCA9800DGKR

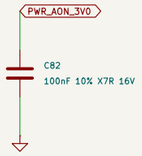


TCA9800 B side: NO external/internal MCU pull-ups.
A-side pull-ups are on switched +3V3. EN follows +3V3.
Debug VTREF is sense only; never power from debugger.



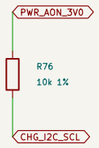


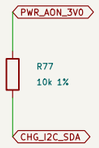


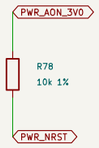


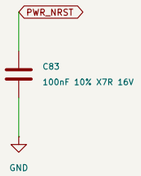
PA0: dedicated active-high charge arm. Reset analog/high-Z
allows R79 at Q8 to hold charging disabled independently of U24.
Never retain this output through standby or run bootloader while armed.

I2C1: PB8/PB7 AF6 -> charger 0x6A + expander 0x41.
I2C2: remap PA9/PA10 to PA11/PA12, AF6.
Host slave address 0x42 reserved; protocol not implemented.









TP2
PWR_AON_3V0

TP3
GND

TP4
PWR_SWDIO

TP5
PWR_SWCLK

TP6
PWR_NRST

Engineering draft / named nets are common / NC = intentional unused pin

Circuit review in progress — not released for fabrication

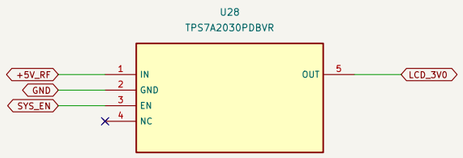
Sheet: /Always-on charging supervisor/
File: power-supervisor.kicad_sch

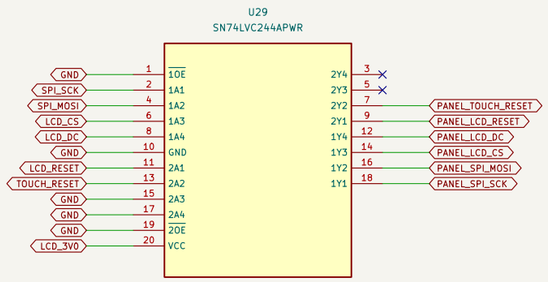
Title: Always-on charging supervisor

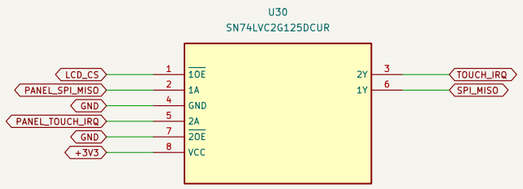
Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 23/25

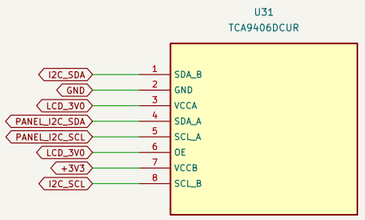
Display supply and signal translation

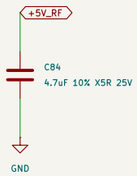
Interface routed and continuity checked. Qualify loads, timing, startup and case cable. Do not fabricate.

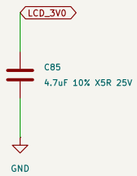


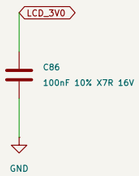


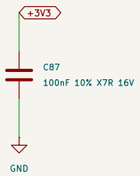


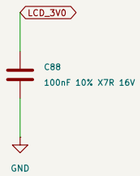


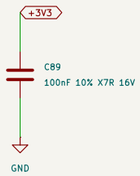


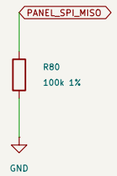


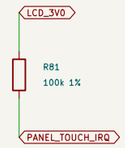












U29: 3.3 V tolerant inputs; outputs use LCD_3V0.
U30: MISO OE follows host LCD_CS; IRQ buffer always enabled.
U31: internal pull-ups: verify bus capacitance and touch option.
U28: 5 V input provides headroom; qualify load and heat.

Engineering draft / named nets are common / NC = intentional unused pin

Circuit review in progress — not released for fabrication

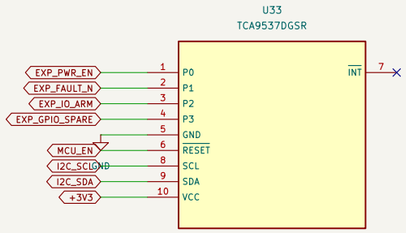
Sheet: /Display supply and translation/
File: display-interface.kicad_sch

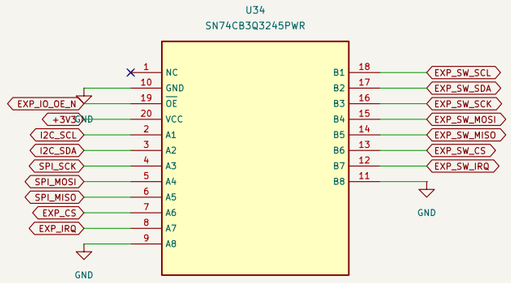
Title: Display supply and signal translation

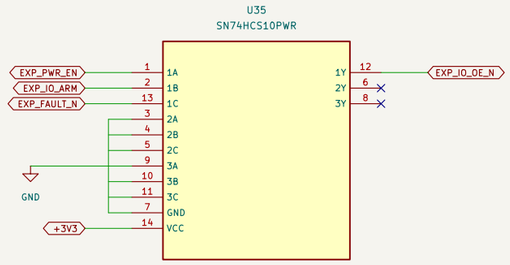
Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 24/25

Expansion power control and signal isolation

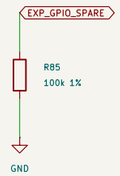
TCA9537 0x49: P0 power, P1 fault, P2 IO arm, P3 unused input. MCU_EN low resets all ports to inputs. External pull-downs retain OFF.

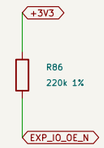














U35: /OE = NOT (EXP_PWR_EN AND EXP_IO_ARM AND EXP_FAULT_N).
Initialize output latch to 0xF0 before direction 0xFA. Power-only 0xF1: armed 0xF5.
Fault is polled; no MCU GPIO/strap changes. A stuck host I2C bus requires hardware reset.
No rail-good sensor; qualify load/startup and share buses only after settling with fault clear.
Accessory address 0x49 is reserved. Swap modules only with handset power off.

Engineering draft / named nets are common / NC = intentional unused pin

Circuit review in progress — not released for fabrication

Sheet: /Expansion control and isolation/
File: expansion-control.kicad_sch

Title: Expansion power control and signal isolation

Size: A3	Date: 2026-09-24	**Rev: A — engineering draft**
KiCad E.D.A. 10.0.6		Id: 25/25